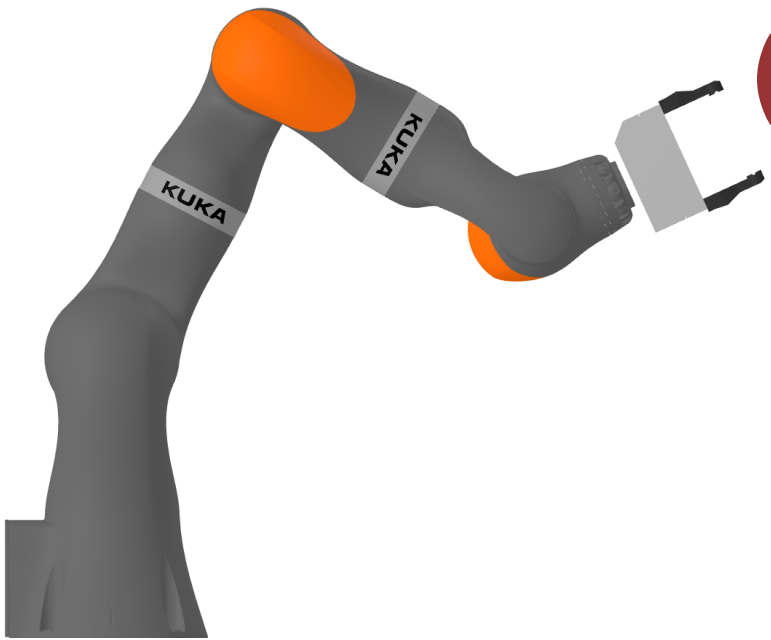
Basic Pick and Place (part 1)



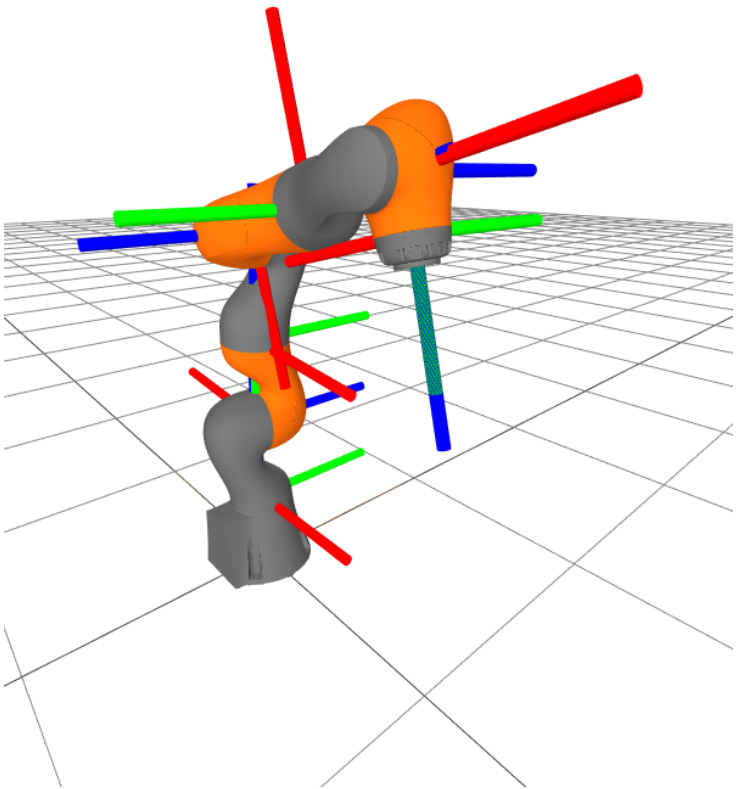
MIT 6.4210/2: Robotic Manipulation

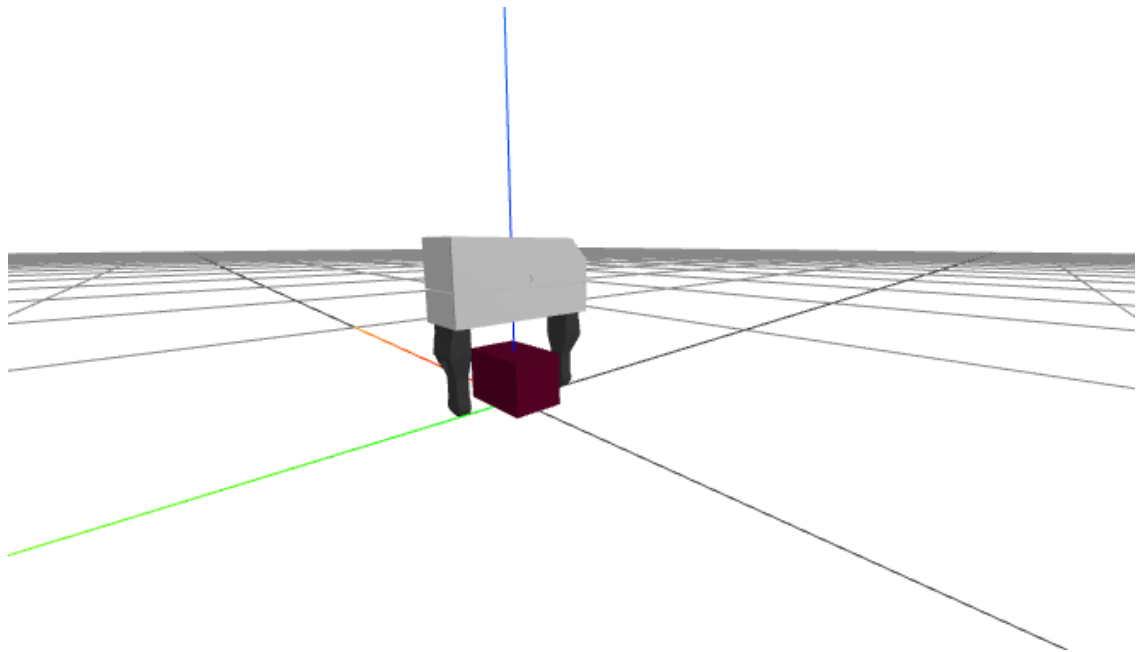
Fall 2025, Lecture 3

Massachusetts Institute of Technology

Basic Pick and Place

Step 1: Kinematic Frames / Spatial Algebra

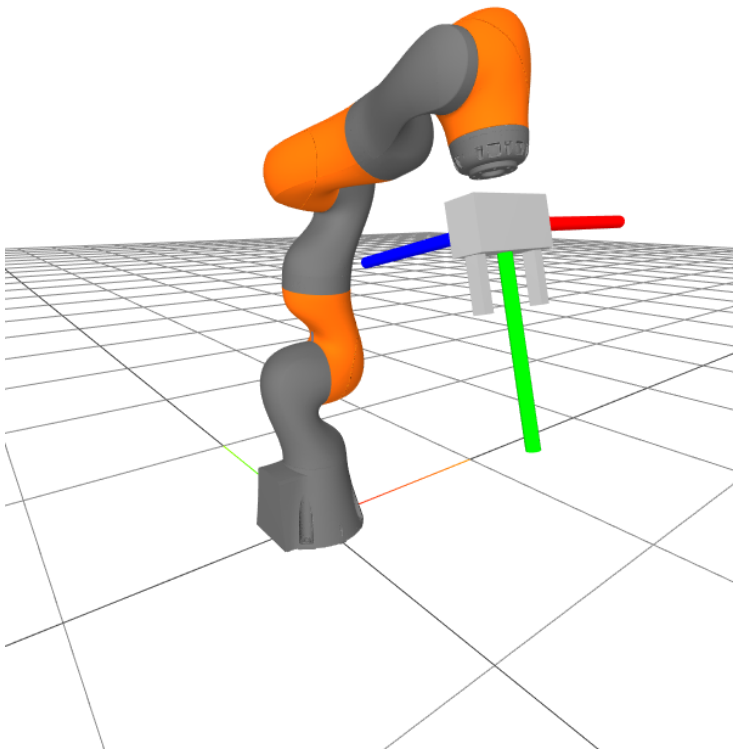


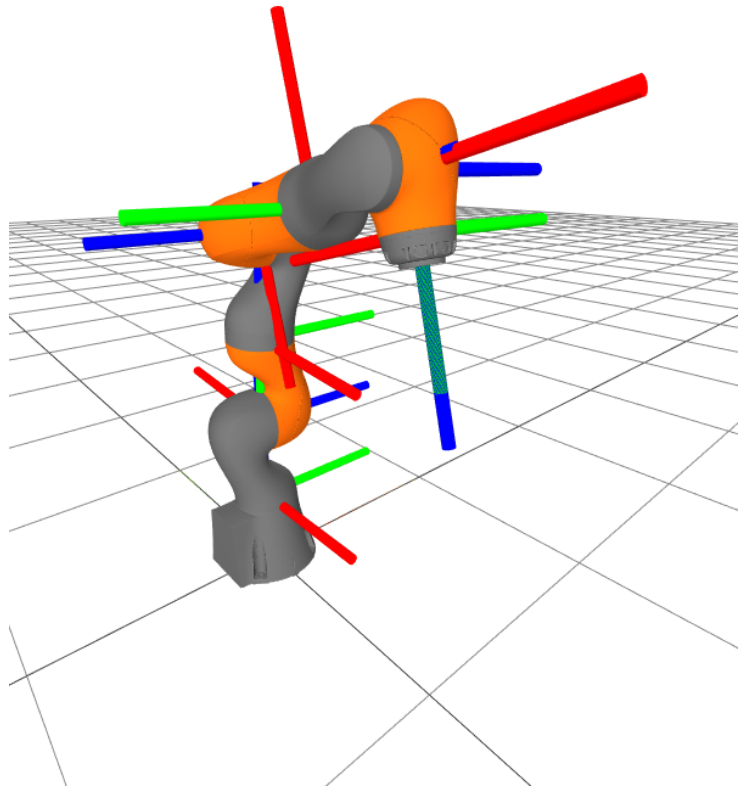


Step 2: Gripper Frame Plan "Sketch"

https://manipulation.mit.edu/data/pick_sketch.html

Step 3: Forward kinematics of the iiwa + WSG





Step 4: Differential Inverse Kinematics

Kinematic Jacobian: $^{W}V^{G} = J^{G}(q)v$

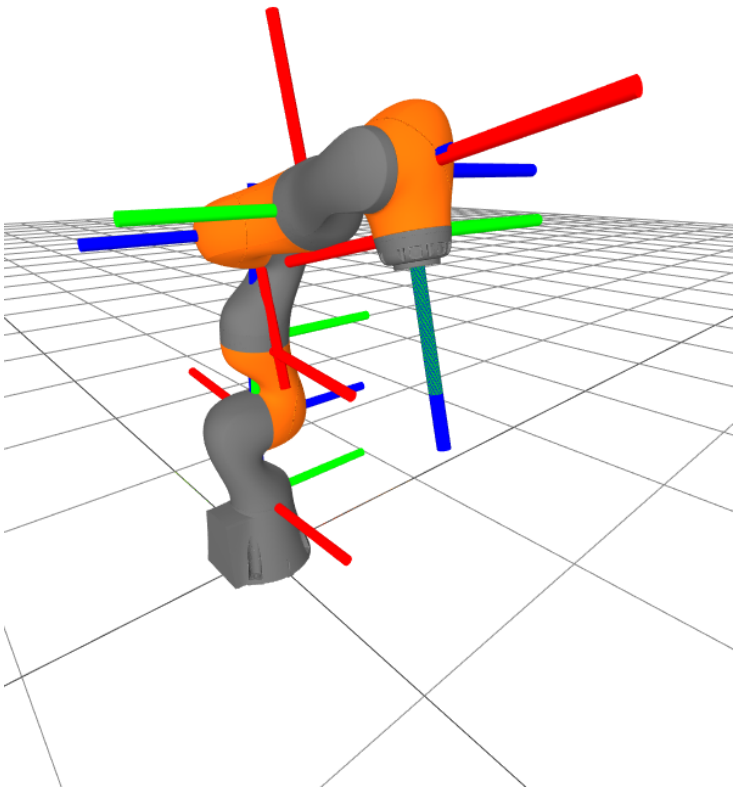
spatial velocity

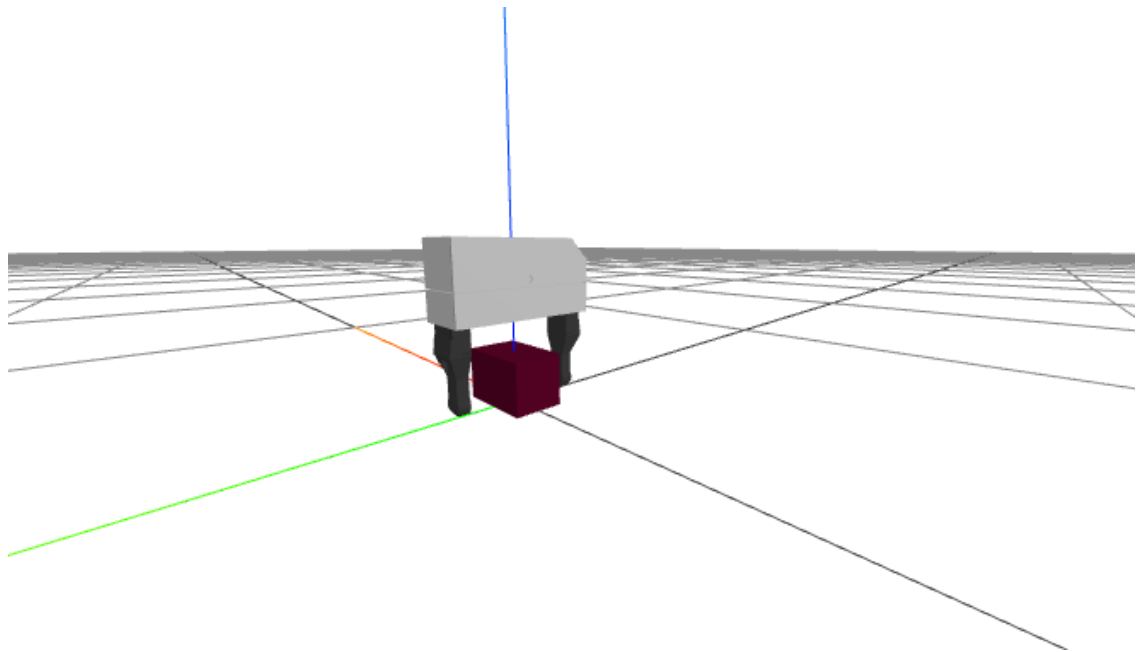
generalized velocity



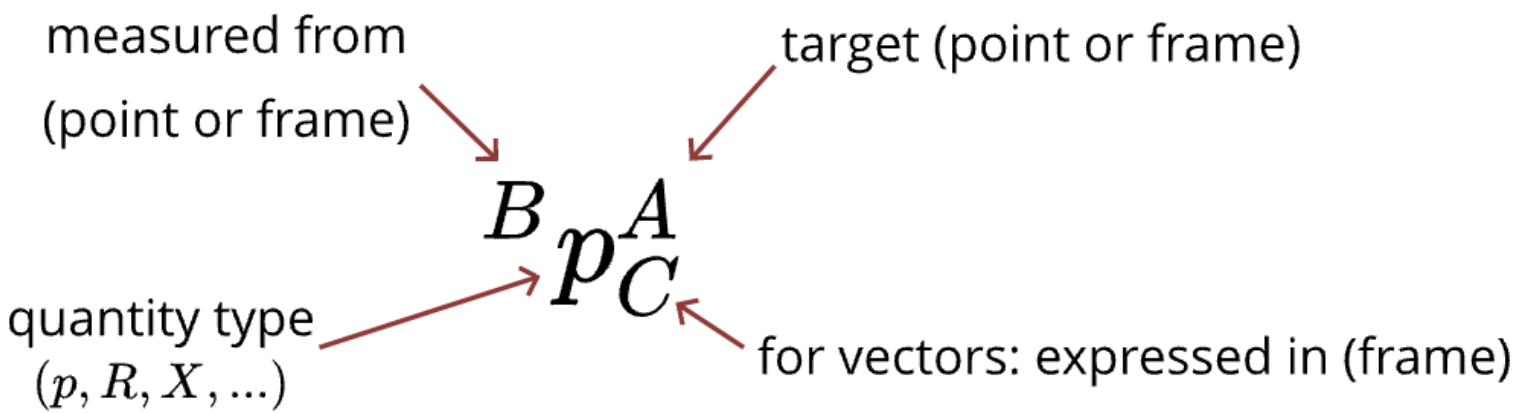
$$v = [J^{G}(q)]^{-1\,W}V^{G}$$

Step 1: Kinematic Frames / Spatial Algebra





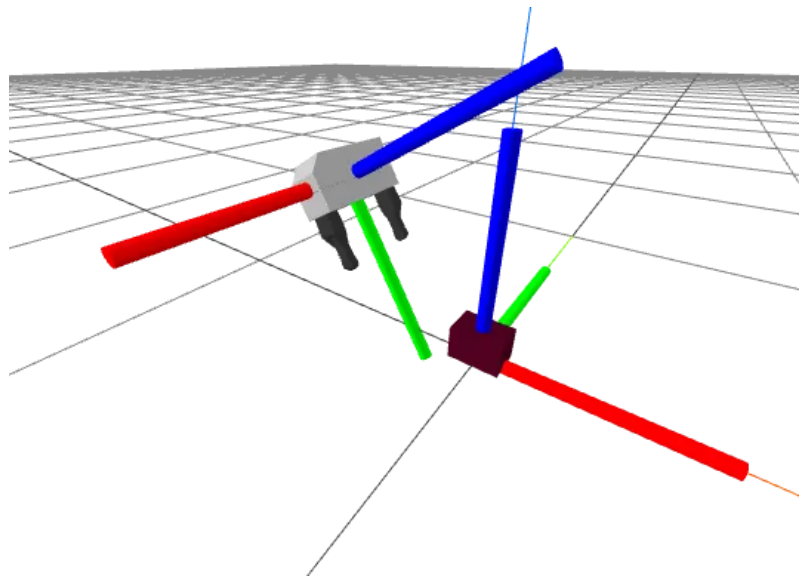
Monogram notation



Check yourself

- G is the gripper frame
- O is the object frame

which is a possible value for ${}^{G}p^{O}$?



a) [.2, 0, -.2]

b) [0, .3, .1]

c) [0, -.3, .1]

Answer: (b)

Check yourself 2

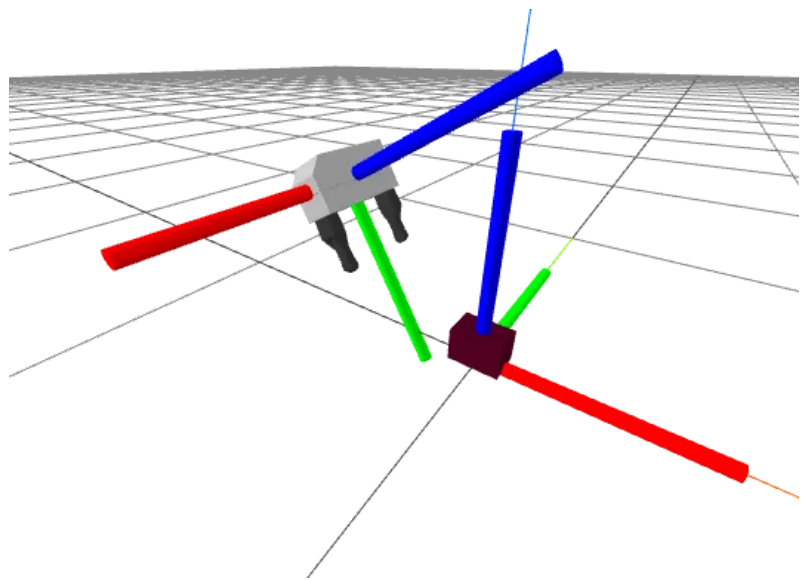
- G is the gripper frame
- O is the object frame

which is a possible value for ${}^{G}p_{O}^{O}$?

a) [.2, 0, -.2]

b) [0, .3, .1]

c) [0, -.3, .1]

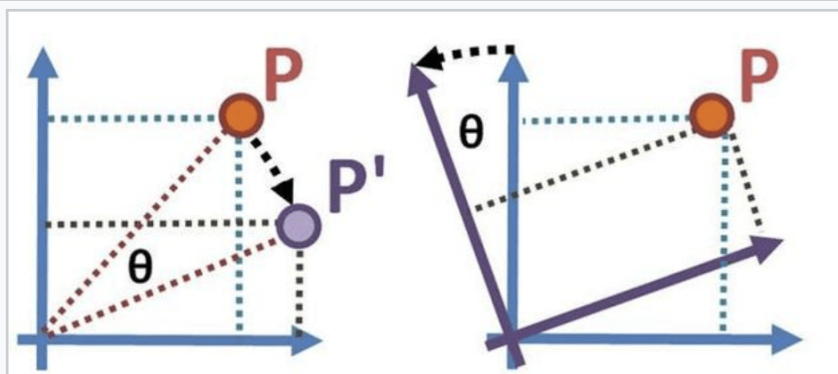


Answer: (a)

https://manipulation.mit.edu/pick.html#spatial_algebra

https://manipulation.csail.mit.edu/pick.html#3D_rotation

Two interpretations of rotations/transforms



In the active transformation (left), a point P is transformed to point P' by rotating clockwise by angle θ about the origin of a fixed coordinate system. In the passive transformation (right), point P stays fixed, while the coordinate system rotates counterclockwise by an angle θ about its origin. The coordinates of P' after the active transformation relative to the original coordinate system are the same as the coordinates of P relative to the rotated coordinate system.

Rotations

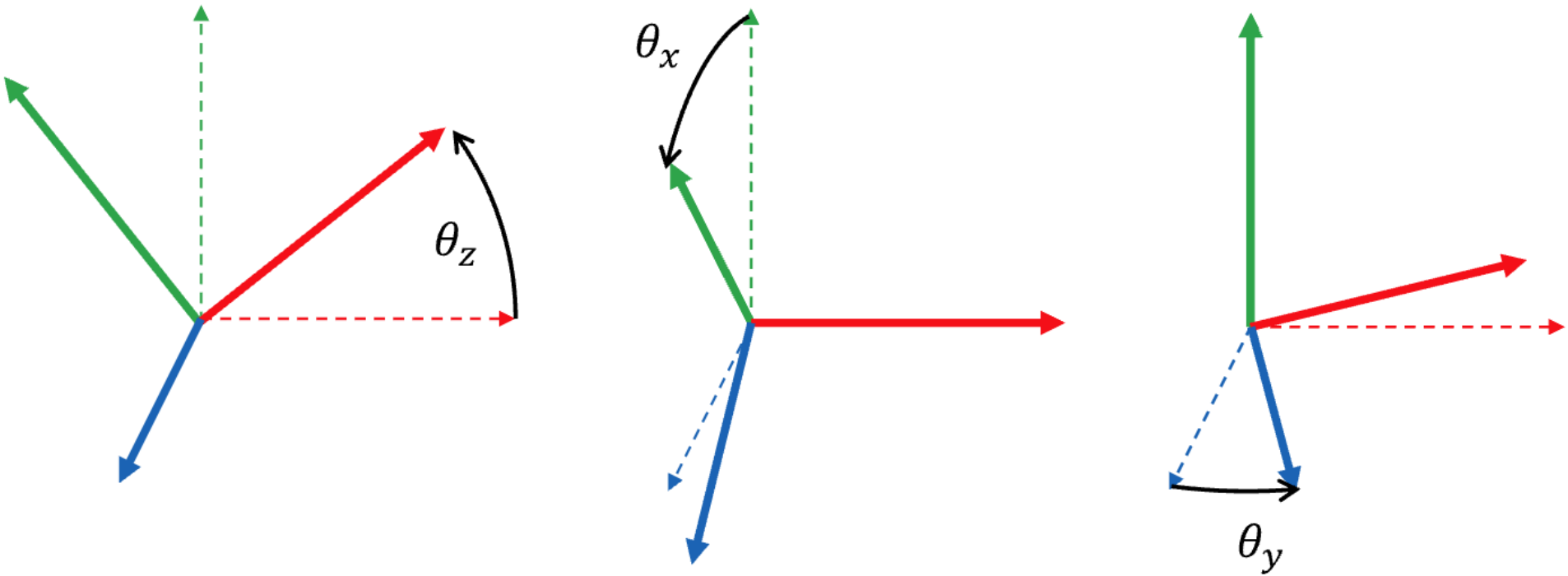
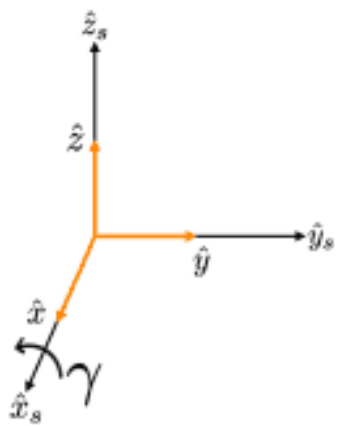
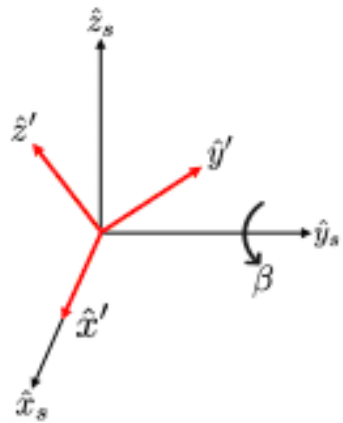


Figure 2. Axis-aligned rotations.

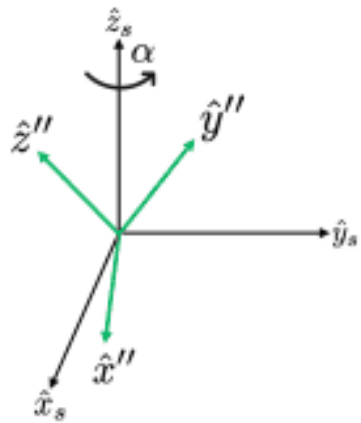
Roll Pitch Yaw (extrinsic)



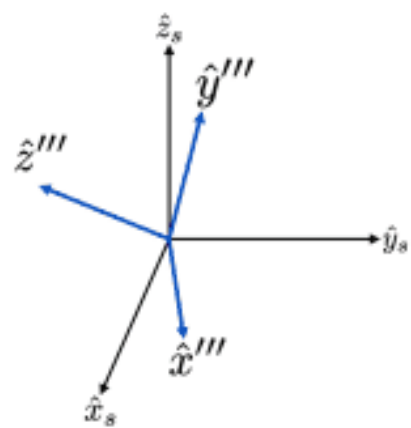
{b} is initially coincident with {s}



frame achieved after a rotation about space frame's x-axis by γ



frame achieved after a rotation about space frame's y-axis by β



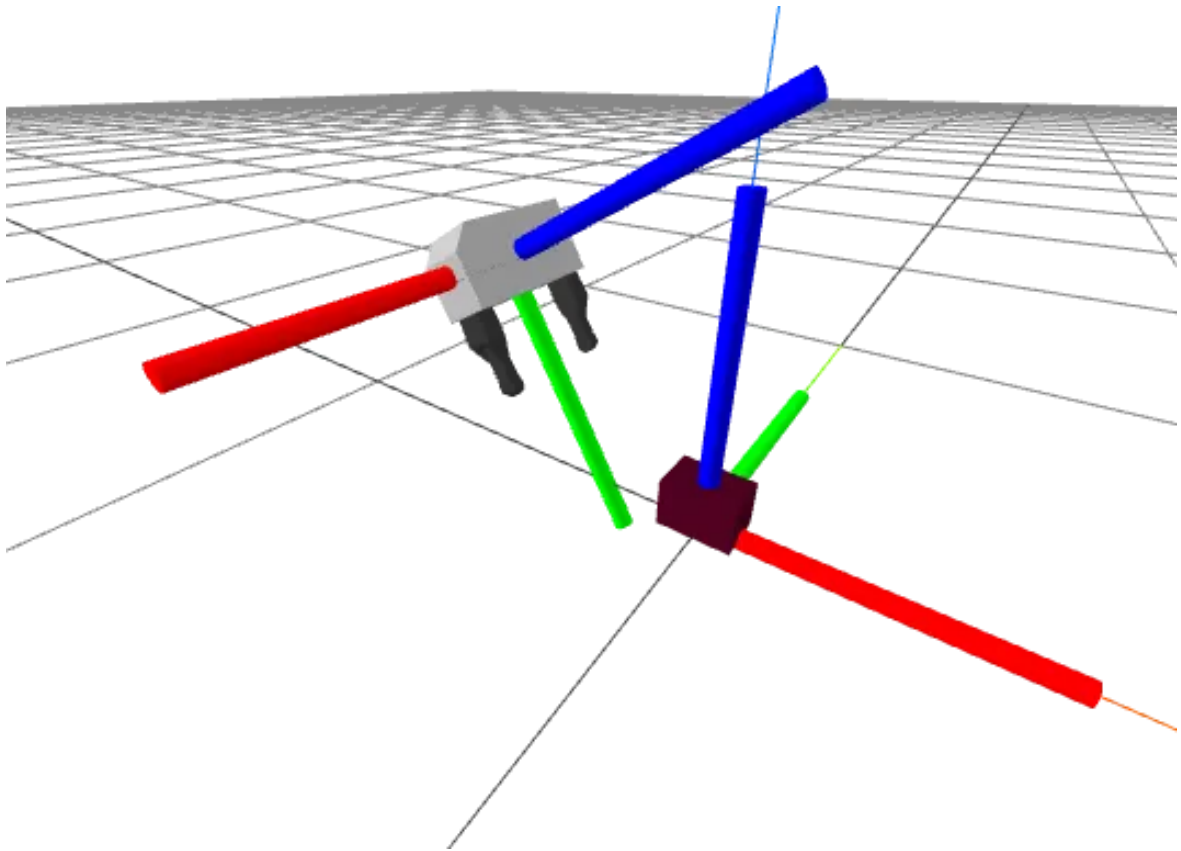
frame achieved after a rotation about space frame's z-axis by α

Mecharithm

Example: Gimbal Lock

Step 2: Gripper Frame Plan "Sketch"

https://manipulation.mit.edu/data/pick_sketch.html



```
def MakeGripperFrames(X_WG, X_WO):
    """
    Takes a partial specification with X_G["initial"] and X_O["initial"] and
    X_O["goal"], and returns a X_G and times with all of the pick and place
    frames populated.
    """
    # Define (again) the gripper pose relative to the object when in grasp.
    p_GgraspO = [0, 0.11, 0]
    R_GgraspO = RotationMatrix.MakeXRotation(
        np.pi / 2.0
    ) @ RotationMatrix.MakeZRotation(np.pi / 2.0)
    X_GgraspO = RigidTransform(R_GgraspO, p_GgraspO)

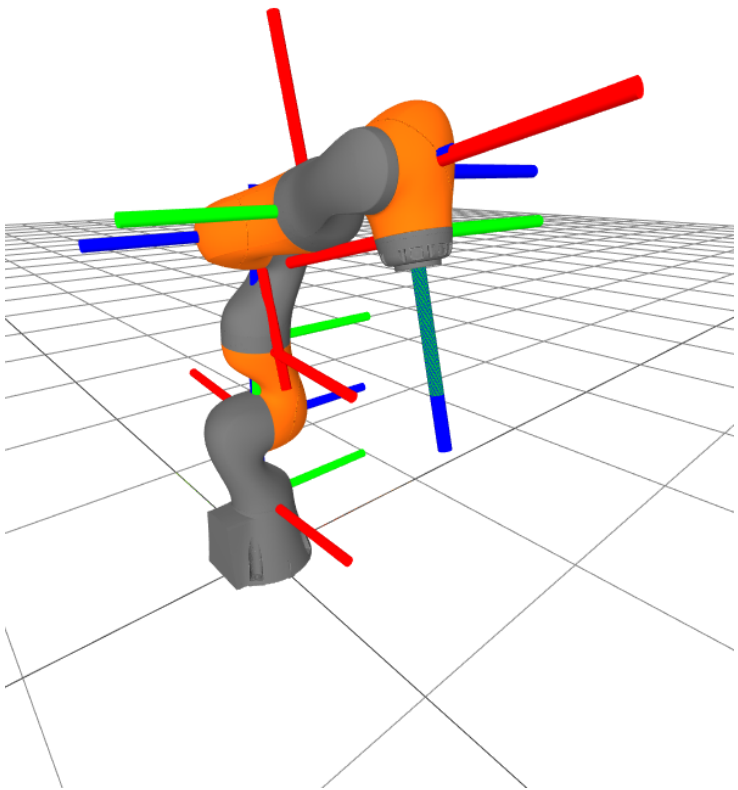
```

Make sure your trajectory safely clears the bins...

https://manipulation.mit.edu/data/pick_fail.html

https://drake.mit.edu/doxygen_cxx/namespacedrake_1_1trajectories.html

Forward Kinematics



$$X^B = f^B_{kin}(q)$$

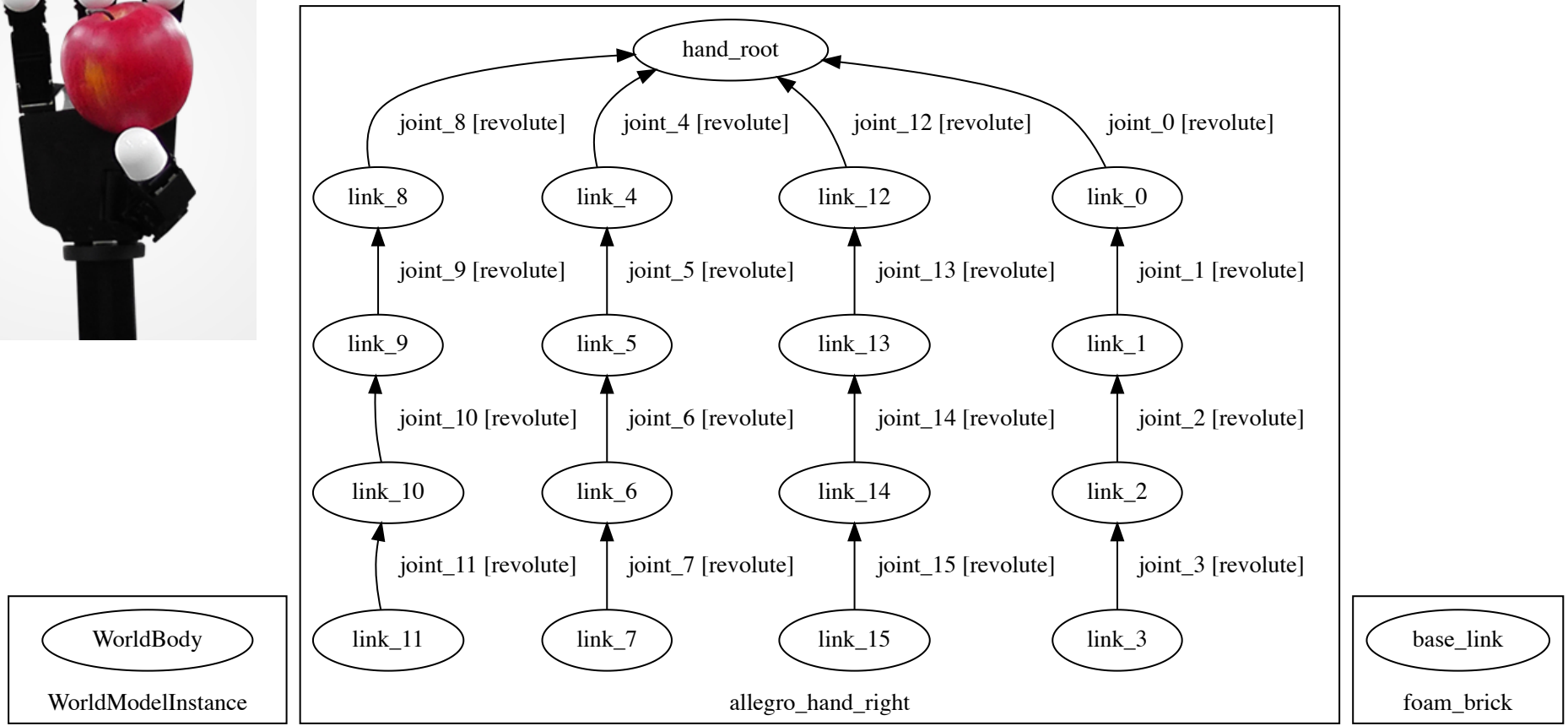
Forward Kinematics

$$ {}^{P}\boldsymbol{X}^{C}(q) = {}^{P}\boldsymbol{X}^{J_P} {}^{J_P}\boldsymbol{X}^{J_C}(q) {}^{J_C}\boldsymbol{X}^{C}. $$

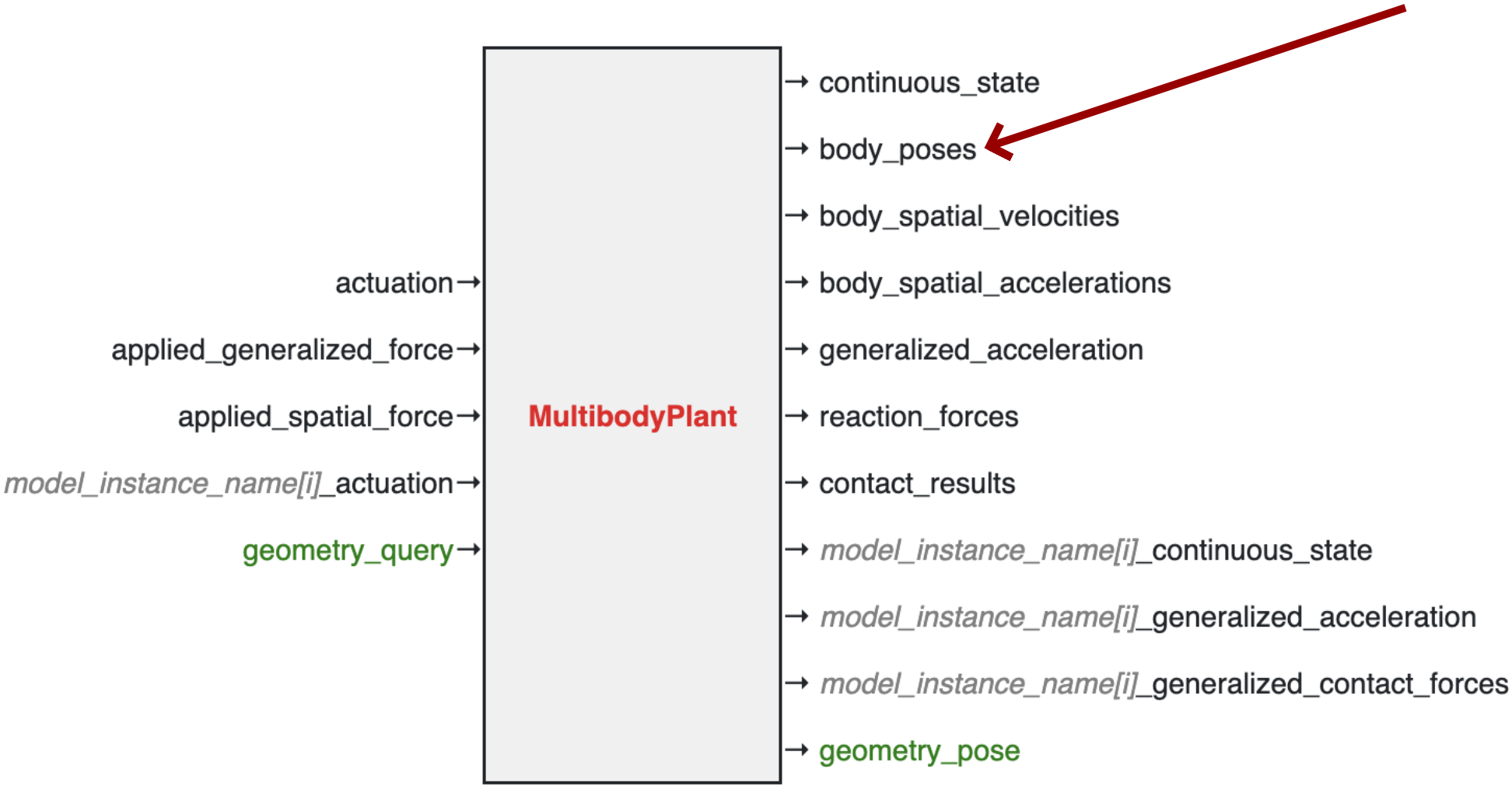
Robot Description Files

```
<?xml version="1.0"?>
<sdf version="1.7">
  <model name="iiwa14">
    <link name="iiwa_link_0">
      <inertial>
        <pose>-0.1 0 0.07 0 0 0</pose>
        <mass>5</mass>
        <inertia>
          <ixx>0.05</ixx>
          <ixy>0</ixy>
          <ixz>0</ixz>
          <iyy>0.06</iyy>
          <iyz>0</iyz>
          <izz>0.03</izz>
        </inertia>
      </inertial>
      <visual name="iiwa_link_0_fixed_joint_lump__iiwa_link_0_visual">
        <geometry>
          <mesh>
            <scale>1 1 1</scale>
            <uri>../meshes/visual/link_0.obj</uri>
```




hand_root
joint_8 [revolute]
joint_4 [revolute]
joint_12 [revolute]
joint_0 [revolute]
link_8
link_4
link_12
link_0
joint_9 [revolute]
joint_5 [revolute]
joint_13 [revolute]
joint_1 [revolute]
link_9
link_5
link_13
link_1
joint_10 [revolute]
joint_6 [revolute]
joint_14 [revolute]
joint_2 [revolute]
link_10
link_6
link_14
link_2
joint_11 [revolute]
joint_7 [revolute]
joint_15 [revolute]
joint_3 [revolute]
link_11
link_7
link_15
link_3
allegro_hand_right
WorldBody
WorldModelInstance
base_link
foam_brick

https://drake.mit.edu/doxygen_cxx/classdrake_1_1multibody_1_1_multibody_plant.html#a6f1fff1ed5cdfefba9e2ce717380783b


actuation
applied_generalized_force
applied_spatial_force
model_instance_name[i]_actuation
geometry_query
MultibodyPlant
continuous_state
body_poses
body_spatial_velocities
body_spatial_accelerations
generalized_acceleration
reaction_forces
contact_results
model_instance_name[i]_continuous_state
model_instance_name[i]_generalized_acceleration
model_instance_name[i]_generalized_contact_forces
geometry_pose

https://drake.mit.edu/doxygen_cxx/classdrake_1_1multibody_1_1_multibody_plant.html#aea0f8aca5f15e90f11e14602b3ba7068

https://manipulation.mit.edu/data/pick.html